Brooks Lab Western Blotting Protocol 01_29_2020

RIPA lysis buffer

Stock concentration	for RIPA buffer, 10ml
100mM Tris-Cl pH 8.0	
300mM NaCl	5ml
10% NP-40 in ddH20	1ml
*10% Na-deoxycholate in ddH20	0.5ml
10% SDS	0.1ml
water	3.4ml

*=light sensitive

Make 100ml of RIPA w/o protease inhibitor, cover, and freeze in 10ml aliquots at -20ºC.
OR
We are using pre-made Pierce RIPA buffer,100ml cat no P189900, which is aliquoted in 10ml aliquots and stored in -20ºC

Before use, thaw and add 1 tablet of protease inhibitor per 10ml aliquot. (Roche complete protease inhibitor cocktail (cat no 04 693 124 001)

I. Harvesting Cells and Preparing Lysate.

*Note***** always keep everything on ice, unless otherwise indicated.*

1. Wash confluent 10 cm plate of cells 2X in ice cold PBS.

2. Add 1 mL cold RIPA with PI to cells, and scrape the cells to remove them from the dish. Transfer to a pre-chilled 2 ml tube on ice.

3. Incubate on ice 20', with periodic vortexing.

4. Pellet the insoluble material by spinning at max speed in refrigerated microcentrifuge 10'.

5. Transfer supernatant to a clean LoBind Protein 1.5 ml tube as lysate. To avoid multiple freeze-thaw cycles, make aliquots (generally 200ul each) *Note****Additional Sonicator or BioEruptor sonication step often needed here for more complete nucleic acid removal. See below.*

6. Store lysates aliquots at -80°C.

II. Sonication of Lysate to break up nucleic acids.

1. Place samples in a 1.5 ml LoBind Protein microfuge tube and prepare a volume balance tube.

2. Bring a timer, ear cuffs and samples and balance on ice to second floor to Kamakaka Lab to get the plastic tube adaptor (in back of second drawer in 2nd bay from the back of the lab.)

3. Sonicator is in cold room down the hall. Put the outflow tube up on the shelf and fill glass reservoir with water to top of white line.

4. Sonicator settings: timer on hold, use max setting (10), constant, then flip on.

5. Hold tubes in the adaptor in the sonicating water bath for 30'.

6. Place tubes on ice to cool for 1 min.

7. Repeat the 30' sonication (=total of two rounds on max at 30'.)

III. BCA Assay for Protein Concentration Determination.

Pierce BCA Protein Assay Kit #23225 or 23227, Thermo Sci with 2mg/ml BSA standard
Use instructions from kit
Equilibrate reagents, samples, and standards to RT. Use the *Microplate reader instructions.*

1. Prepare a dilution series of BSA in working range of 20-2000ug/ml from 1 glass vial of stock 2 mg/ml BSA in the kit for the standard curve:

Vial	RIPA buffer, ul	2mg/ml BSA or standard,ul	Final conc ug/ml
A	0	300 of stock	2000
B	125	375 of stock	1500
C	325	325 of stock	1000
D	175	175 of B	750
E	325	325 of C	500
F	325	325 of E	250
G	325	325 of F	125
H	400	100 of G	25
I	400	0	0

2. Prepare fresh working reagent 50:1 of A:B, enough for all standards, unknowns, and replicates of the unknowns (should have n=3)

3. Add 25ul of unknown or standard to 96-well plate well followed by 200ul of working reagent using 200ul pipettor and gently mix avoiding spillover.

4. Cover plate with foil and incubate 37ºC for 30 min in shaking incubator, gently shaking.

5. Cool to room temp.

6. Measure Absorbance on the VarioSkan at 562nm.

7. Determine protein concentration of your unknown from the standard curve (the Skanit software will plot the curve and the unknowns on it if you edit the standards in the plate layout to include the concentrations above in the table.) Export report as an excel doc.

IV. SDS PAGE:

Prepare the following solutions for SDS-PAGE, Transfer, and Antibody Incubations:

Running Buffer
100ml 10x TGX running buffer
900ml dd water

Transfer Buffer
200ml 5x Biorad Transfer Buffer
200ml 190 proof ethanol
600ml dd water

PBST
50ml 10x PBS
0.5ml Tween-20
bring vol to 500ml with dd water

5% Milk Block (prepare this one immediately before use)
Safeway brand non-fat milk pouch.
5gms in 100ml PBST

Reagents:

-Biorad Mini-PROTEAN Precast Gels TGX 4-15% SDS-PAGE with Tris Glycine (cat no 456-1083) with 10 well capacity, 30ul volume per well.
-Biorad 4x Laemmli Sample Buffer (cat no 161-0747)
-Biorad 2-mercaptoethanol (cat no 161-0710)
-Precision Plus protein standards (cat no 161-0374)

A. Denaturing samples in Laemmli reducing buffer:

1. Add 100ul 2-mercaptoethanol to 900ul 4x Laemmli sample buffer to make MLB.
2. Dilute samples with MLB at a ratio of 3 parts sample to 1 part MLB, so for 30ul final volume, add 7.5ul MLB to 22.5ul of sample in RIPA in LoBind safety lock tube, 1.5ml.
3. Denature at 95ºC/5 minutes.

4. Transfer to ice.

B. Loading and Running Gel:

1. Make 1 Liter of Running Buffer (Tris-Glycine, cat no).

2. Remove gel from package, take off green strip at the bottom, and rinse the wells three times with about a ml of running buffer (use 1 ml pipettor).

3. Place gel, tall plate facing out in outer side of a holder and the reservoir block at the other to make a running buffer reservoir. Put the other gel holder in place to take up space.

4. Fill the chamber to the 2-gel mark and test the circuit by checking for bubbles after turning on power to 70 volts.

5. Load 30ul of samples and 10ul of Biorad Precision Plus protein ladder Dual Color (cat no 161-0374) on gel using 20ul cone-tipped pipettor.

6. Run at 100-120V, depending on desired resolution, until adequate separation of ladder lanes in sizes regions of interest.

7. Photograph the gel after the run with your phone to help keep track of orientation.

V. Semi-dry Transfer to PVDF Membrane.

1. Pre-soak blotting stacks in 1X transfer buffer made according to directions on the bottle.

2. Pre-wet PVDF membrane from Biorad Transblot kit in 190 proof ethanol, then in transfer buffer.

3. Layer into Transblot drawer (stack, membrane, gel, second stack, then roll out bubbles gently with conical tube)

4. Run the transfer on Biorad setting, standard Mininigel TGX, 25mA, 25V, 3 mins.

5. While running, pour 5% milk block in 1X PBST directly into cleaned small black Western blot box (cat no Z742099-5EA).

6. Incubate transferred blot 1h at room temperature in 5% milk block in 1X PBST.

7. Photograph the blot after the transfer to help keep track of orientation.

VII. Antibody Incubations and ECL visualization of target protein bands

Primary Antibodies:

β-Actin (C4): sc-47778 http://datasheets.scbt.com/sc-47778.pdf This is HRP-conjugated. 43

ECL SuperSignal West Pico Chemiluminescent Substrate
(cat no 34077)

A. Primary and secondary antibody binding

Use 1:500 Dilutions for primary antibodies, in 5% milk/PBST
Note: Primary antibody (**unconjugated with HRP**) can be re-used 3x if kept frozen.

1. Drain milk block from blots and add primary Ab.

2. Incubate in cold room over-night covered and shaking on orbital shaker, covered with plastic.

Next day,

1. Wash blot 3x/15min in PBST.

2. Freshly prepare secondary antibody at 1:10,000 dilution (2ul to 20ml 5%milk/PBST).

3. Incubate in secondary antibody for 1h, RT.

4. Discard secondary and wash blot 3x/5 mins in 1xPBST.

5. Place drained blot into black box the size of the blot and add ECL working solution (1ml of each reagent mixed in a 15ml conical and applied to blot.)

6. Incubate in ECL for 5 min, shaking occasionally by hand, covered.

7. Drain blot and place between plastic sheets for imaging using Biorad GelDoc, as below. After imaging, the blot can be probed for actin as a protein loading control without stripping the blot, using anti-actin-HRP as described below.

B. Actin staining with anti-actin-HRP primary

1. Wash blot 3x with 1X PBST< 10 mins.

2. Incubate 20 mins, room temp in the primary actin antibody 1:500 freshly made dilution in 5% Milk block.

3. Wash 2-3x in 1x PBST 5min each.

4. Incubate in ECL as above and image.

C. Imaging Blot on BioRad ChemiDoc XRS+ (Vollmers lab)

1. Place blot in between sheet protector plastic and transport to imager in cassette to keep dark.

2. Place blot on white screen.

3. Place screen into chamber drawer.

4. Select new protocol, then blot, then chem hi sensitivity and position blot under live focus setting.

5. Under Applications, select chemidoc hi sensitivity to photograph target bands. Note the image size and the gain (2X) and bin (2x2) to make sure the image size is the same when the protein standards are photographed in order to be able to merge the images.

6. Under Live Acquire, select acquisition settings: 1, 600, for range and total =100 acquisitions (the 100 can be lowered for longer exposure), starting at 0.25 start time for high expression target.

7. Freeze and save the image of your choice as it comes up.

8. **Without moving the blot**, take a picture to visualize the protein standards by selecting Custom under Applications, then create an Epi illumination protocol with the same gain and binning as used for photographing the bands (double check the image size prediction under the options…)

9. Merge picture in Image Lab software using Image tools.

*Note*****Imaged blots can be stored at 4ºC in PBST for stripping and re-probing. Some folks also store them frozen in -20ºC flat in plastic bag.